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WorldMedQA-V: a multilingual, multimodal medical examination dataset for multimodal language models evaluation

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Abstract

Multimodal/vision language models (VLMs) are increasingly being deployed in healthcare settings worldwide, necessitating robust benchmarks to ensure their safety, efficacy, and fairness. Multiple-choice question and answer (QA) datasets derived from national medical examinations have long served as valuable evaluation tools, but existing datasets are largely text-only and available in a limited subset of languages and countries. To address these challenges, we present WorldMedQA-V, an updated multilingual, multimodal benchmarking dataset designed to evaluate VLMs in healthcare. WorldMedQA-V includes 568 labeled multiple-choice QAs paired with 568 medical images from four countries (Brazil, Israel, Japan, and Spain), covering original languages and validated English translations by native clinicians, respectively. Baseline performance for common open- and closed-source models are provided in the local language and English translations, and with and without images provided to the model. The WorldMedQA-V benchmark aims to better match AI systems to the diverse healthcare environments in which they are deployed, fostering more equitable, effective, and representative applications.[1]

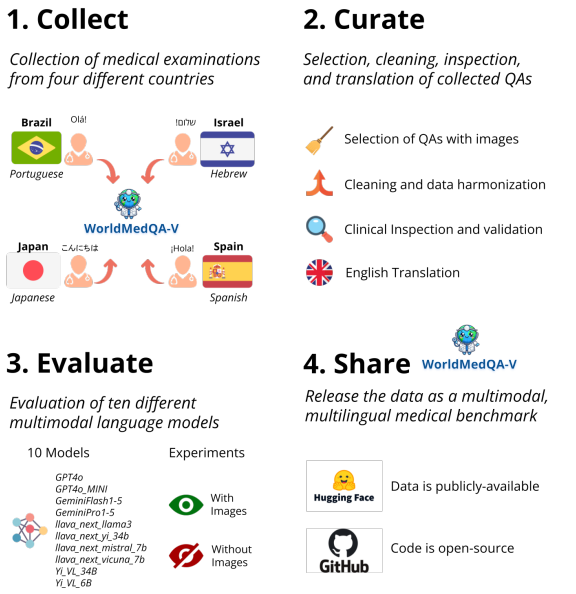

Figure 1: WorldMedQA-V dataset generation and evaluation workflows.

1 Introduction

Generative artificial intelligence (AI) models are increasingly being adopted in healthcare, highlighting the need for robust benchmarks to assess their safety, efficacy, and fairness (Thirunavukarasu et al., 2023; Clusmann et al., 2023; Abbasian et al., 2024; Wiggers, 2024).

One of the key evaluation tasks in Natural Language Processing (NLP) is Question Answering (QA)(Yu et al., 2024; Fan et al., 2023), which involves building systems that can automatically respond to human queries in natural language by combining language understanding with information retrieval (Jin et al., 2020). Multi-choice QA benchmarks have become essential not only for evaluating large language models (LLMs) but also for assessing vis language models (VLMs) in medicine (Liu et al., 2024).

Recent research has explored the performance of LLMs in medical exams, with ChatGPT being the first AI system to pass the USMLE (Kung et al., 2023), prompting further studies (Gobira et al., 2023; Liu et al., 2024; Chen et al., 2024b). A recent review identified 45 studies on ChatGPT's performance in medical exams (Liu et al., 2024), but VLMs remain underexplored in medical tasks (Yan et al., 2023; Wu et al., 2023). Despite progress, current models face limitations such as context fragility, biases, and inconsistent multilingual per-

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[1]All code is accessible on https://github.com/WorldMedQA/V and the dataset on https://huggingface.com/datasets/WorldMedQA/V.

formance (Gallifant et al., 2024; Zack et al., 2024; Chen et al., 2024a). There is also a need for more diverse datasets to ensure equitable AI evaluation in healthcare (Restrepo et al., 2024b). Key gaps include:

- **Real-world validity**: Studies reveal errors in existing medical QA datasets (Saab et al., 2024).
- **Linguistic diversity**: Many datasets lack language representation (Appendix Table 1) (Restrepo et al., 2024a,b; Ryan et al., 2024).
- **Imaging data**: Most medical QA benchmarks exclude multimodal data (Appendix Table 1)
- **Training data contamination**: Outdated datasets may overlap with LLM/VLM training corpora (Zhang et al., 2024a,b; Gallifant et al., 2024).

To address these issues, we introduce `WorldMedQA-V`, a multilingual, multimodal dataset for evaluating language and vision models. Key contributions include:

- **Multimodal medical exams from four countries**, supporting local languages and English.
- Previously **unseen** multimodal exam questions with **clinical validation** by medical professionals.
- Baseline **performance reporting of current state-of-the-art VLMs** across languages, including an evaluation of performance differentials between **local languages and English**.
- An investigation into the **impact of adding image data** to model performance and **stability across language** translations.

2 Related Work

Recent benchmarks like MMMU (Zhang et al., 2023b), EXAMS-V (Zhang et al., 2023a), and CulturalVQA (Wang et al., 2023) evaluate VLMs across multiple languages and disciplines, revealing notable performance gaps across linguistic and cultural contexts. Studies show that VLMs perform better in English, likely due to the predominance of English training data (Adam et al., 2023; Weidinger et al., 2021). These findings highlight the need for improving VLMs in diverse languages and cultural settings, especially in specialized domains.

Appendix A.1 Table 1 summarizes existing medical QA datasets by country. Six languages are covered, spanning seven countries across three continents: Asia (China, India, South Korea, and Taiwan), Europe (Spain and Sweden), and North America (U.S.).

Medical datasets from these regions highlight challenges in LLMs' performance in healthcare. In Asia, notable datasets include those from China (Li et al., 2021a), Taiwan (Jin et al., 2020), South Korea (Kweon et al., 2024), and India (Pal et al., 2022). The MLEC-QA dataset from China, with 136,236 multiple-choice questions, is the largest. Despite LLMs being pre-trained on vast datasets, performance in this domain is hindered by limited diversity and quality of training data, especially for two-step reasoning and biomedical concepts (Li et al., 2021a). Similar trends are observed in Taiwan and South Korea, where English-pretrained models underperform on local medical exams. In Europe, datasets from Spain, Sweden, and Poland (the latter not publicly available) underscore the difficulties LLMs face, especially as question complexity increases (Vilares and Gómez-Rodríguez, 2019a). However, recent advancements saw models like *GPT3.5-Turbo* and *GPT4* pass the Swedish medical licensing exam (Hertzberg and Lokrantz, 2024a), while *GPT4-Turbo* slightly outperformed humans in Poland (Bean et al., 2024).

3 Methodology

Figure 1 shows the overall workflow of the study.

3.1 Data Collection

Our study uses medical exam data from Brazil, Israel, Japan, and Spain, consisting of multiple-choice questions from national licensing or specialization exams. Brazil's dataset includes 100 questions per exam from the 2011–16 and 2020–24 "Revalida" exams. Israel's dataset contains 150 questions from Phase A of the resident certification exam (2020–23). Japan's data comes from the 116th–118th National Medical Licensing Examinations (2022–24), while Spain's dataset includes questions from specialization exams (2019–23). Further details are provided in Appendix A.2.

3.2 Clinical Validation

A clinical validation process was carried out for all collected and translated data to ensure their quality and relevance. Native-speaking clinicians from

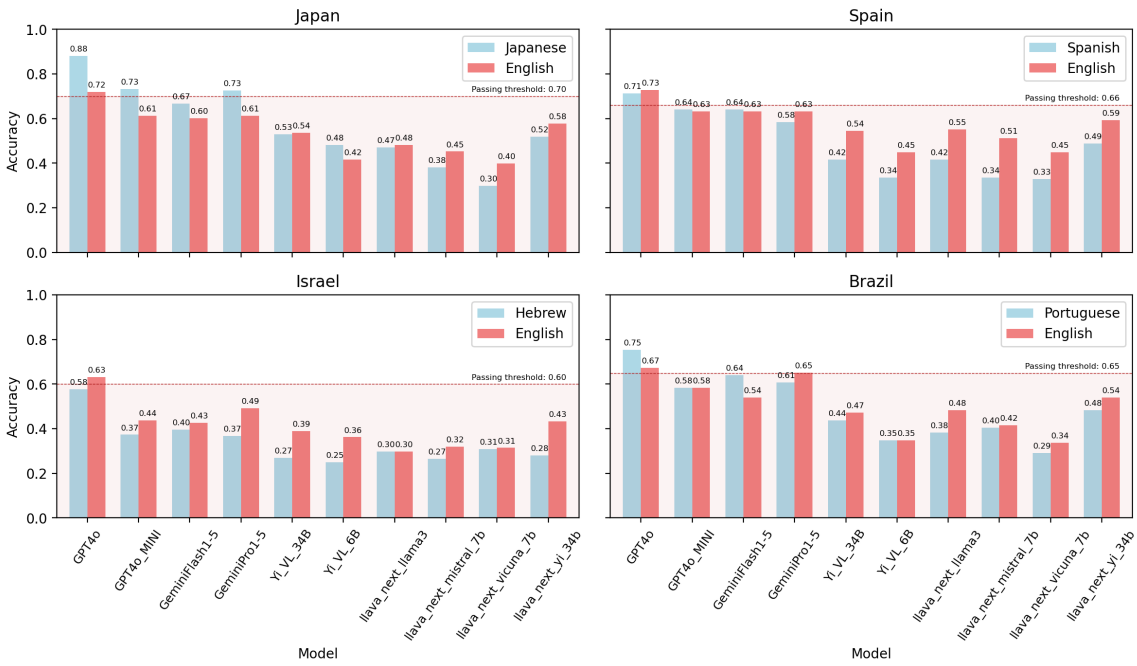

Figure 2: Accuracy in local language and English across models and countries. The red-shaded area highlights each country's exam passing threshold. Passing score is a proxy here since our dataset is a subset. Detailed results in Appendix A.6

each country validated the three key stages of the process — data extraction, translation, and final QA review.

3.3 Evaluation

Models: We included open- and closed-source models across a range of sizes: *GPT4o-2024-05-13*, *GPT4o-MINI-2024-07-18*, *GeminiFlash1-5 May*, *GeminiPro1-5 May*, *llava-next-llama3(8B)*, *llava-next-yi-34b*, *llava-next-mistral-7b*, *llava-next-vicuna-7b*, *Yi-VL-34B*, and *Yi-VL-6B*. All models were set to generate 512 tokens, with a temperature of 0 for reproducibility, and evaluated with Nvidia-GPU with CUDA > 12.0.

Experiments: The *VLMEvalKit* evaluation framework (Duan et al., 2024) was utilized to conduct experiments. We evaluated the ten models with and without image input, using accuracy as the metric. Cohen's kappa coefficients (Cohen, 1968) were computed to assess each model's reliability when answering the question in the original language versus the English translation.

4 Results and Discussion

Dataset: The complete `WorldMedQA-V` includes a total of 726 QAs and 850 images across four countries: Brazil, Israel, Japan, and Spain. Each QA is paired with at least one image, though some images appear in more than one question. After the exclusion of questions with multiple images or correct options, the final evaluation subset contains 568 QAs, each with a single associated image and correct option. Table 2, in Appendix A.3, provides a detailed summary of data distribution across countries and languages. Box 1 in Appendix A.4 shows an example from the Brazilian dataset.

VLMs' Performance: Figure 2 shows model performance across datasets in the local language and in English. Compared to the previously reported performance of *GPT4* on the USMLE, which is 90% (Brin et al., 2023), all models exhibit reduced performance when confronted with both image and text data. *GPT4o* emerged as the best-performing model. The only dataset for which *GPT4o* did not achieve a passing grade was the Israel dataset in Hebrew on which it achieved only 58%. Interestingly, *GPT4o* passed the English-translated version (63%) of the Israeli dataset. The other dataset in `WorldMedQA-V` with a non-Roman alphabet is the Japan dataset, on which *GPT4o* achieved an accuracy of 88%, exceeding the 70% passing threshold. This may be because Japanese is better represented in pretraining datasets and has character overlap with Mandarin. The underperformance in Hebrew, in contrast, could reflect Hebrew's lower representation in pretraining data,

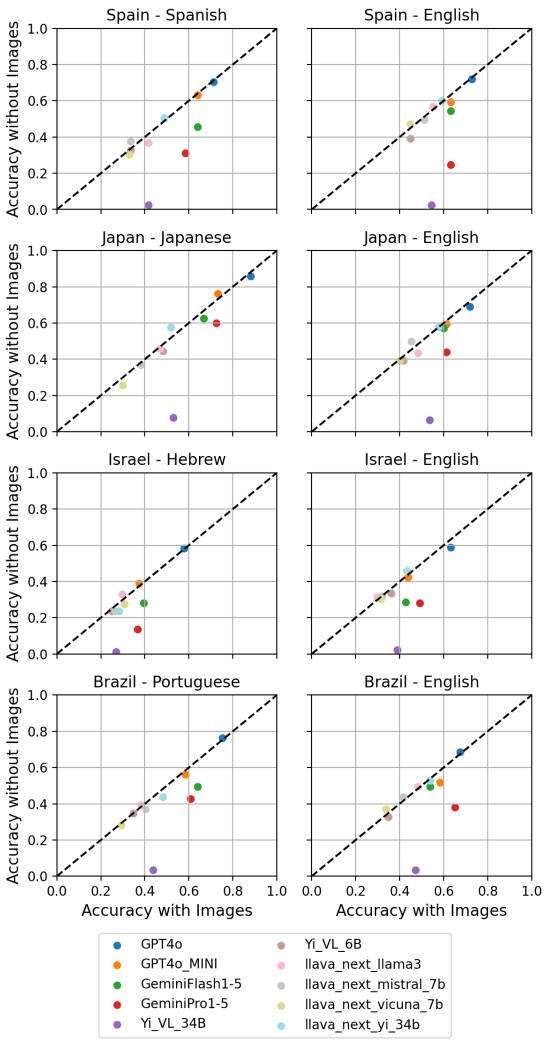

Figure 3: Accuracy across countries and languages with and without image input. See Appendix A.6 for details.

affecting the models' ability to understand the native language as effectively (Üstün et al., 2024). Models generally performed better on English-translated datasets, particularly for the Spain and Israel datasets. Moreover, the English-translated Israel dataset exhibits a somewhat lower overall performance when compared to other countries, which may indicate data variations that go beyond language. In the Brazilian subset, *GPT4o* scored 75% in Portuguese and 67% in English. Similarly, in the Japanese dataset, models such as *GPT4o*, *GPT4o-MINI*, *GeminiFlash1-5*, and *GeminiPro1-5* performed better in Japanese than in English, indicating strong language support for Japanese. The lowest accuracies were from the *llava-next* series, particularly on the Israel dataset, where several variants achieved nearly random accuracies in both Hebrew and English, ranging 24-46%. (Figure 2)

Accuracy with and without image input: Models performed better with image input. This trend was consistent across most datasets, particularly for models with lower baseline performance. However, the accuracy of the *GPT* models showed only minor variations — typically within 1-3% — regardless of whether the image was provided. Models from the *Gemini* family tended to be most sensitive to the exclusion of images, with improvements ranging from 4-27% when images were provided. The *Yi-VL* and *llava-next* models, which generally underperformed across the board, exhibited more stochastic variations in either direction depending on image input. Lastly, it is worth noting that the *Yi-VL-34b* model had almost no predictive power without images. (Figure 3)

Model consistency comparing English and local languages: Table 4 in Appendix A.5 compares model outputs in original languages to English translations using Cohen's kappa. *GPT4o* consistently achieved the highest agreement, particularly in the Brazil, Japan, and Spain datasets, with better performance in image-based settings. The highest kappa (84%) was observed in Spain's text-only setting, likely due to the model's high overall accuracy. Models like *GPT4o-MINI* and *GeminiFlash1-5* performed well in Brazil and Spain but lagged behind *GPT4o*. In contrast, *Yi-VL* showed lower agreement across countries, suggesting worse cross-language consistency. Notably, *GeminiPro1-5* showed an improvement in kappa, from 16.3% to 69.3%, when images were included in the Spanish set, demonstrating a substantial stabilizing effect of multimodal input. Overall, model cross-linguistic consistency improved with image data input.

5 Conclusion

In this work, we introduce `WorldMedQA-V`, a clinically validated, multilingual, and multimodal dataset containing medical QAs and images from Brazil, Israel, Japan, and Spain. We evaluated the performance of 10 VLMs across both local languages and English translations, highlighting performance disparities between languages and the role of multimodal data in enhancing model accuracy. Our results show that while most models performed better with image input, language diversity remains a critical challenge, particularly for underrepresented languages like Hebrew. `WorldMedQA-V` addresses key gaps in language diversity and multimodal evaluation in healthcare, providing a robust benchmark for future VLM development.

6 Limitations

While WorldMedQA-V represents a significant step toward creating a multilingual, multimodal benchmark for evaluating VLMs in healthcare, several limitations must be acknowledged.

First, the dataset, while carefully curated by trained physicians to ensure the validity of both questions and answers, remains relatively small. As we evaluated 568 multiple-choice questions and images, the sample size is limited in comparison to larger text-based benchmarks.

Second, the dataset only includes data from four countries: Brazil, Israel, Japan, and Spain, spanning three continents. This geographic limitation results in an underrepresentation of certain regions, particularly Africa, North and Central America, Oceania, and other parts of Asia.

Furthermore, although the benchmark introduces multimodal elements, it pairs only one image per question. Real-world clinical scenarios often involve multiple images from different time points or modalities, such as a sequence of X-rays, CT scans, and pathology slides. Another limitation is that text that is within images were not translated or adapted. English translations, although validated by native-speaking clinicians from each country, require further cross-validations, as these are typically nontrivial tasks.

Additionally, the lack of open-source multimodal medical language models restricts our ability to comprehensively evaluate and compare state-of-the-art health AI using WorldMedQA-V. Furthermore, since the models we tested were not originally trained for the medical domain, some LLMs (e.g., Gemini) refused to respond when no image was provided for certain questions, resulting in lower scores. When evaluating model performance against a passing threshold, a limitation is that our analysis relies on a limited set of multiple-choice questions with images, which may not provide consistent difficulty levels across different questions within the same exam.

Lastly, we set the underlying assumption that each question had only one correct answer, excluding cases where multiple correct answers were possible. This decision was made to simplify evaluation, but it may not reflect the inherent ambiguity and complexity found in both medical examinations and real-world medical scenarios where multiple treatment options or diagnoses can be valid.

Acknowledgments

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A Appendix

A.1 Existing publicly-available medical examination QA dataset per country

Table 1: Summary of existing open-source Medical QA Datasets by Country.

Country	Dataset	#QA	Language(s)	Modalities	Source	Years
China	MedQA (Jin et al., 2020)	34,251	Simplified Chinese	Text	MCMLE, Mainland China Medical Licensing Examination	Not Clear
China	MLEC-QA (Li et al., 2021b)	136,236	Simplified Chinese	Text, Images	National Medical Licensing Examination (NMLEC)	Not Clear
India	MedMCQA (Pal et al., 2022)	193,155	English	Text	AIIMS PG, NEET PG	1991-2022
Spain	Head-QA (Vilares and Gómez-Rodríguez, 2019b)	6,765	Spanish and English	Text, Images	Ministerio de Sanidad, Consumo y Bienestar Social	2013–2017
Republic of Korea	KorMedMCQA (Kweon et al., 2024)	5,345	Korean and English	Text	Korea Health Personnel Licensing Examination Institute	2012–2023
Sweden	MedQA-SWE (Hertzberg and Lokrantz, 2024b)	3,180	Swedish	Text	National Board of Health and Welfare, Umeå University	2016–2023
Taiwan	MedQA (Jin et al., 2020)	14,123	Traditional Chinese	Text	TWMLE, Taiwan Medical Licensing Examination	Not Clear
United States	MedQA (Jin et al., 2020)	12,723	English	Text	USMLE, United States Medical Licensing Examination	Not Clear

A.2 Details on collected data per country

A.2.1 Brazil

The examination data were collected from the "Revalida" examinations, which are publicly available on the Brazilian government's official website. The "Revalida" exam, administered by the National Institute of Educational Research and Studies (INEP), supports the process of diploma revalidation for doctors who graduated abroad and wish to practice in Brazil. The exams consist of two sections: 100 multiple-choice questions (20 in each of the following areas: Internal Medicine, Surgery, Pediatrics, Preventive Medicine, and Gynecology and Obstetrics) and open-ended questions. For this work, only the multiple-choice section was included. Data from the years 2011–2016 and 2020–2024 were used, encompassing all publicly available years at the time of this study (de Estudos e Pesquisas Educacionais Anísio Teixeira | Inep).

A.2.2 Israel

The Israeli subset consists of questions from seven medical specialties: Internal Medicine, Clinical Microbiology, Neurology, Oncology, Ophthalmology, Urology, and Public Health. These questions are drawn from Phase A of the two-phase examination process that residents in Israel must complete during their training. Phase A is a written exam held annually, comprising approximately 150 questions. We included questions from tests administered between 2020 and 2023, with full versions of these exams publicly available on the Israel Medical Association's website (Association; Association).

The questions are categorized into three main types: preclinical cases, clinical cases, and questions based on scientific articles. Preclinical cases focus on foundational scientific knowledge, while clinical cases present patient background information followed by clinical questions related to the patient's medical conditions. Questions derived from scientific articles involve analysis of graphs, figures, and study results, which are particularly prevalent in the public health exam. Some questions also include visual aids, such as diagnostic images, laboratory slides (e.g., blood smear slides in Clinical Microbiology), and other data specific to patients' clinical presentations.

A.2.3 Japan

Japanese questions were sourced from past examinations that were published on the website of the Ministry of Health, Labour and Welfare. We included the 116th, 117th, and 118th National Medical Licensing Examination (Japanese Ministry of Health Labour and Welfare 2022; Japanese Ministry of Health Labour and Welfare 2023; Japanese Ministry of Health Labour and Welfare 2024), which corresponded to 2022-2024.

A.2.4 Spain

The examination data were sourced from the annual exams organized by the Ministerio de Sanidad, Consumo y Bienestar Social (Spanish Ministry of Health, Consumer Affairs, and Social Welfare) (de Sanidad). These exams are part of the competitive selection process for specialized medical positions in Spain's public healthcare system. Eligibility for participation requires candidates to possess a bachelor's degree in medicine (6 years of study) and typically prepare for a year or more, given the limited number of vacancies. The exams play a critical role in ranking candidates, who are able to select their specialization and hospital placement only based on their exam performance. For this study, only data from the years 2019–2023 were used to avoid overlap with the existing Head-QA dataset (Vilares and Gómez-Rodríguez, 2019b).

A.3 Detailed Data Statistics

Table 2: `WorldMedQA's` data across countries and languages. In the curated dataset, each QA was associated with at least one image. Some images were present in more than one question. In the final subset for evaluation (rightmost column), each question had a single image and the correct option associated with it, resulting in fewer samples. The number of answer options per question (fourth column) refers to the original number of choices in the multiple-choice format (e.g., A-D for four options or A-E for five options). However, all questions after preprocessing results in 4 options only. In cases where this varies, such as in Brazil, the value represents a weighted average across questions. The total number of QAs and images does not immediately add up due to some questions sharing images or having multiple associated options.

Country	**Language**	**Years**	**Option/QA**	**QAs, n (%)**	**Images, n (%)**	**Final, n (%)**
Brazil	Portuguese	2011-2024	4.27	93 (12.8%)	94 (11.1%)	89 (15.7%)
Israel	Hebrew	2020-2023	4.00	200 (27.6%)	184 (21.6%)	186 (32.7%)
Japan	Japanese	2022-2024	5.00	306 (42.1%)	445 (52.4%)	168 (29.6%)
Spain	Spanish	2019-2023	4.00	127 (17.5%)	127 (14.9%)	125 (22.0%)
Total	4 Languages	2011-2024	4.00	726 (100%)	850 (100%)	568 (100%)

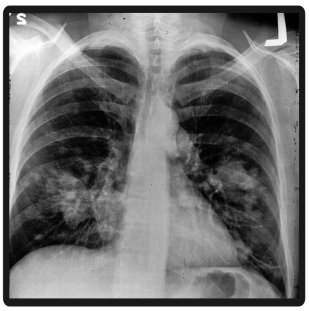
A.4 Example QA from the Brazilian dataset

Box 1. Example multimodal QA from the Brazilian subset

Original (Portuguese)
Um paciente do sexo masculino, 55 anos de idade, tabagista 60 maços/ano, com tosse crônica há mais de 10 anos, relata que há cerca de três meses observou a presença de sangue na secreção eliminada com a tosse. Refere ainda perda de cerca de 15% do peso habitual nesse mesmo período, anorexia, adinamia e sudorese noturna. A radiografia de tórax realizada por ocasião da consulta é mostrada abaixo. Qual a hipótese diagnóstica mais provável nesse caso?

A) Aspergilose pulmonar.
B) Carcinoma pulmonar.
C) Tuberculose cavitária.
D) Bronquiectasia com infecção.
E) Doença pulmonar obstrutiva crônica.

Image



Translation (English)
A 55-year-old male patient, with a smoking history of 60 pack-years, has had a chronic cough for over 10 years. He reports that about three months ago, he noticed the presence of blood in the sputum. He also mentions a weight loss of about 15% of his usual weight during the same period, anorexia, weakness, and night sweats. The chest X-ray taken at the time of the consultation is shown below. What is the most likely diagnostic hypothesis in this case?

A) Pulmonary aspergillosis.
B) Lung carcinoma.
C) Cavitary tuberculosis.
D) Bronchiectasis with infection.
E) Chronic obstructive pulmonary disease.

A.5 Model output consistency across countries and test setting

Table 3: Cohen's Kappa reflecting agreement between languages for the same models, countries, and testing setting. Values in **bold** highlight the model with highest kappa per country and testing mode. The two studied settings were text-only (T. only) and text and image (T. & I.).

Country	**Brazil**		**Israel**		**Japan**		**Spain**	
Model ↓ Setting →	**T. only**	**T. & I.**	**T. only**	**T. & I.**	**T. only**	**T. & I.**	**T. only**	**T. & I.**
GPT4o	**0.684**	**0.743**	**0.619**	0.654	**0.683**	**0.618**	**0.840**	**0.829**
GPT4o-MINI	0.642	0.655	0.458	0.603	0.525	0.554	0.715	0.809
GeminiFlash1-5	0.612	0.536	0.533	**0.655**	0.591	0.521	0.594	0.767
GeminiPro1-5	0.389	0.469	0.184	0.416	0.351	0.490	0.163	0.693
Yi-VL-34B	0.020	0.438	0.111	0.309	0.033	0.393	0.030	0.507
Yi-VL-6B	0.427	0.320	0.150	0.204	0.240	0.348	0.269	0.251
llava-next-llama3	0.429	0.441	0.401	0.380	0.269	0.264	0.435	0.433
llava-next-mistral-7b	0.498	0.310	0.148	0.243	0.153	0.234	0.466	0.348
llava-next-vicuna-7b	0.385	0.491	0.167	0.281	0.091	0.185	0.310	0.279
llava-next-yi-34b	0.592	0.635	0.208	0.223	0.393	0.373	0.594	0.488

A.6 Detailed performance with and without images

Table 4: Accuracy comparison across countries and **original** languages (Portuguese, Hebrew, Japanese, and Spanish) for each model. The two studied settings were text-only (T. only) and text and image (T. & I.). Each cell represents the performance of each model in its native language dataset, highlighting how the presence or absence of images affects accuracy.

Country	**Brazil**		**Israel**		**Japan**		**Spain**	
Model ↓ Setting →	**T. only**	**T. & I.**	**T. only**	**T. & I.**	**T. only**	**T. & I.**	**T. only**	**T. & I.**
GPT4o	0.764	0.753	0.584	0.578	0.857	0.881	0.704	0.712
GPT4o-MINI	0.562	0.584	0.389	0.373	0.762	0.732	0.632	0.640
GeminiFlash1-5	0.494	0.640	0.281	0.395	0.625	0.667	0.456	0.640
GeminiPro1-5	0.427	0.607	0.135	0.368	0.601	0.726	0.312	0.584
Yi-VL-34B	0.034	0.438	0.011	0.270	0.077	0.530	0.024	0.416
Yi-VL-6B	0.348	0.348	0.238	0.249	0.446	0.482	0.328	0.336
llava-next-llama3	0.393	0.382	0.330	0.297	0.458	0.470	0.368	0.416
llava-next-mistral-7b	0.371	0.404	0.238	0.265	0.369	0.381	0.376	0.336
llava-next-vicuna-7b	0.281	0.292	0.276	0.308	0.256	0.298	0.304	0.328
llava-next-yi-34b	0.438	0.483	0.238	0.281	0.577	0.518	0.504	0.488

Table 5: Accuracy comparison across countries and **English-translated** datasets for each model. The two studied settings were text-only (T. only) and text and image (T. & I.). Each cell represents the performance of each model after translation, highlighting how the presence or absence of images affects accuracy.

Country	**Brazil**		**Israel**		**Japan**		**Spain**	
Model ↓ Setting →	**T. only**	**T. & I.**	**T. only**	**T. & I.**	**T. only**	**T. & I.**	**T. only**	**T. & I.**
GPT4o	0.685	0.674	0.589	0.632	0.690	0.720	0.720	0.728
GPT4o-MINI	0.517	0.584	0.422	0.438	0.595	0.613	0.592	0.632
GeminiFlash1-5	0.494	0.539	0.286	0.427	0.571	0.601	0.544	0.632
GeminiPro1-5	0.382	0.652	0.281	0.492	0.440	0.613	0.248	0.632
Yi-VL-34B	0.034	0.472	0.022	0.389	0.065	0.536	0.024	0.544
Yi-VL-6B	0.326	0.348	0.335	0.362	0.393	0.417	0.392	0.448
llava-next-llama3	0.494	0.483	0.314	0.297	0.435	0.482	0.568	0.552
llava-next-mistral-7b	0.438	0.416	0.319	0.319	0.500	0.452	0.496	0.512
llava-next-vicuna-7b	0.371	0.337	0.303	0.314	0.393	0.399	0.472	0.448
llava-next-yi-34b	0.528	0.539	0.459	0.432	0.577	0.577	0.600	0.592